Technology Stack

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit Buddy – AI Fitness Plan Generator using Gemini Models
Maximum mark	2marks

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	Frontend / Client-Side	HTML, CSS, Jinja2	Used to create a simple, responsive and user-friendly interface for entering fitness details and displaying workout plans.
2	Backend / Server-Side	Python, Fast API	Fast API handles user requests, API routes and communication between the frontend, database and Gemini AI.
3	Database / Data Storage	SQLite, SQL Alchemy	SQLite stores user details and workout plans, while SQL Alchemy provides easy database management using Python.
4	Cloud / Hosting / Deployment	Unicorn (Local Deployment)	Unicorn runs the Fast API application efficiently during development and testing.
5	Version Control & CI/CD	Git, GitHub	Used for version control, project backup, code

			management and collaboration.
6	Third-Party APIs / Other Tools	Google Gemini API	Gemini AI generates personalized 7-day workout plans and nutrition/recovery suggestions based on user inputs.

Summary

Fit buddy uses **HTML, CSS and Jinja2** for the frontend, **Python and Fast API** for the backend, **SQLite and SQL Alchemy** for data storage, **Unicorn** for running the application, **Git/GitHub** for version control, and the **Google Gemini API** for AI-powered fitness plan generation.